Grocery Ordering App — Technical Overview

A visual walkthrough of the application (frontend screens) and how information moves through it.

> **Please read first.** This app is **self-contained**: it runs on a phone (or in a web browser) and does **not** talk to any outside server, website, database, or artificial-intelligence service. Everything it needs to work lives **inside the app itself**.
>
> Normally an app has a "backend" — a separate computer on the internet that stores data and does the heavy lifting. This app doesn't. Instead, it has a small **"engine" built into the app** that plays that role: it remembers your data, does the calculations (like cart totals), and even *pretends* a delivery is on its way. The sections below explain that built-in engine in plain language. Everything shown is real and taken from the actual code — nothing has been made up.

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1. Application Overview

A grocery ordering app for **vegetables & fruits** with two roles:

- **Customer** — browse/search products, build a custom package, add to cart, checkout, and watch **simulated live delivery tracking**.
- **Admin** — an analytics dashboard plus CRUD management of products, packages and orders. Admin order/delivery-status changes flow **live** to the customer's tracking screen through shared state.

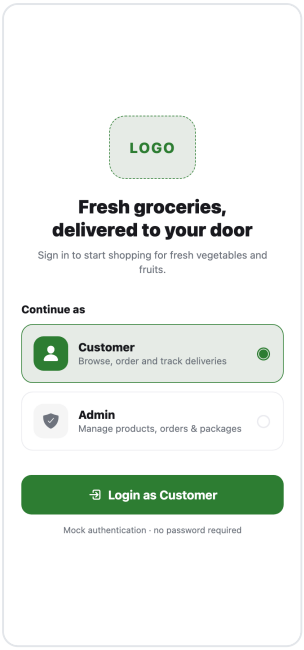
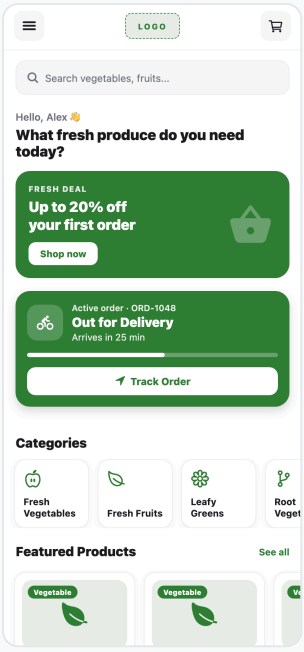
Stack: React Native 0.81 · Expo SDK 54 · TypeScript · React Navigation (drawer + native stack) · `react-native-chart-kit` / `react-native-svg` for analytics · AsyncStorage for persistence.

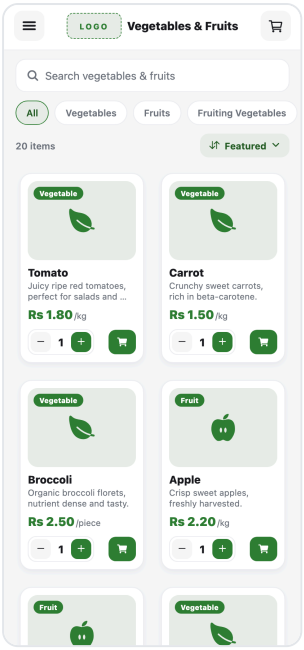
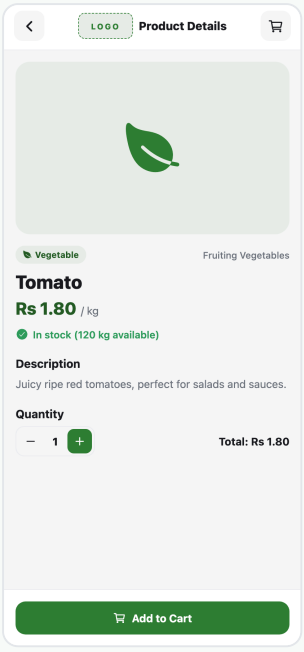
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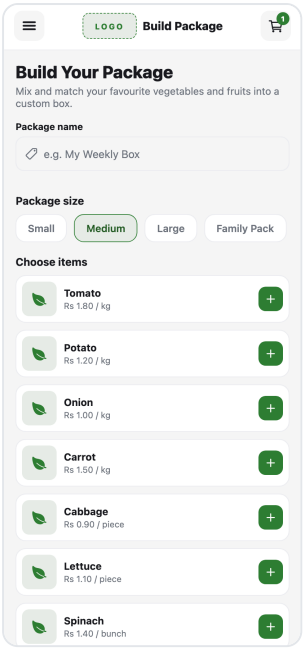
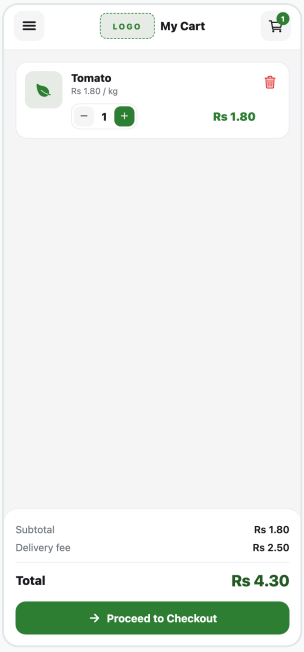
2. Frontend Screens

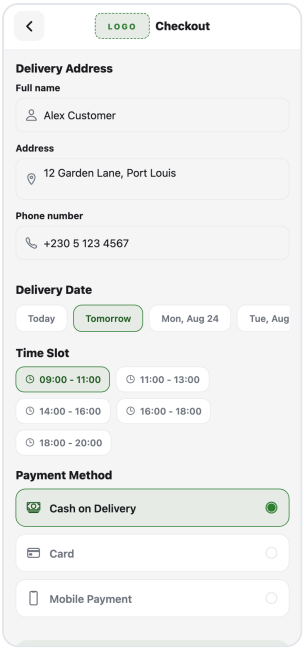
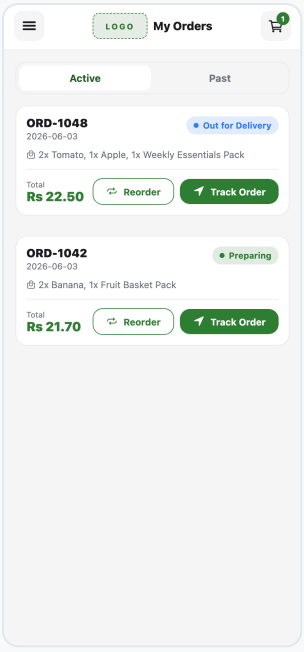
All images below are real screenshots captured from the app running locally via `npx expo start --web`.

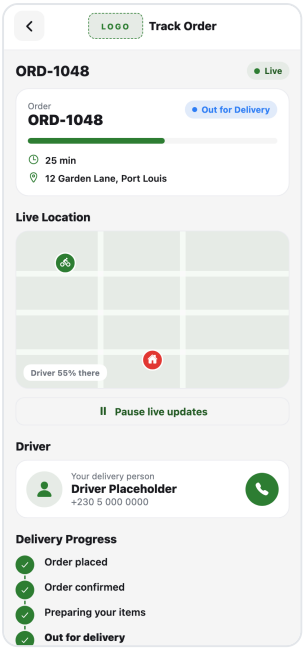
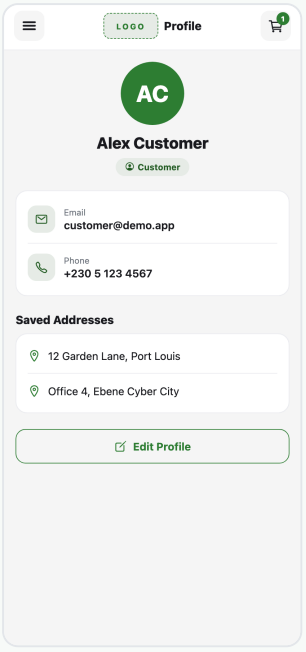
Customer experience

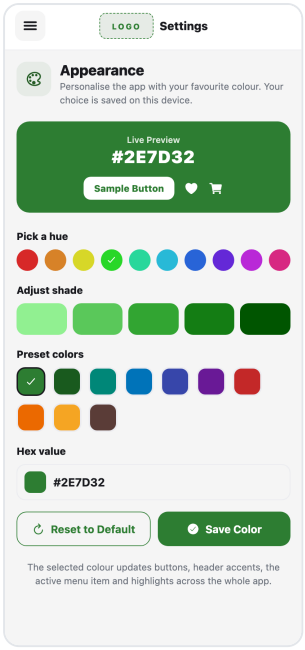
Screen	What it does
	**Login** — mock, password-less auth. Pick **Customer** or **Admin** to enter the matching experience.
	**Home** — search bar, promo banner, a **live active-order card** with progress + "Track Order", category shortcuts and featured products.

Screen	What it does
	**Vegetables & Fruits** — searchable, category-filtered, sortable product grid with inline quantity + add-to-cart.
	**Product Details** — image, price/unit, stock status, description, quantity selector and running total, with **Add to Cart**.

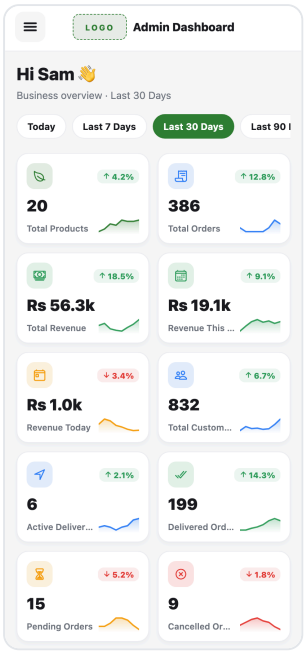
Screen	What it does
	**Build Your Package** — compose a custom box (name + size + chosen items with quantities) as a single cart line.
	**My Cart** — line items with quantity controls, subtotal, fixed delivery fee and total, then **Proceed to Checkout**.

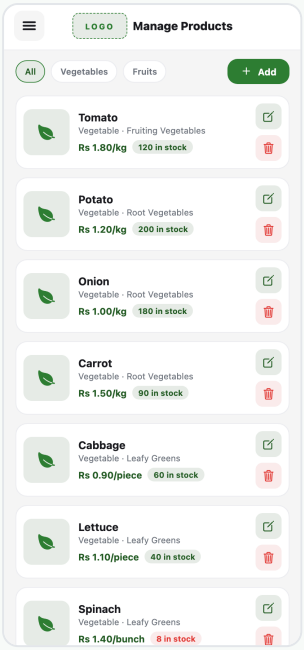
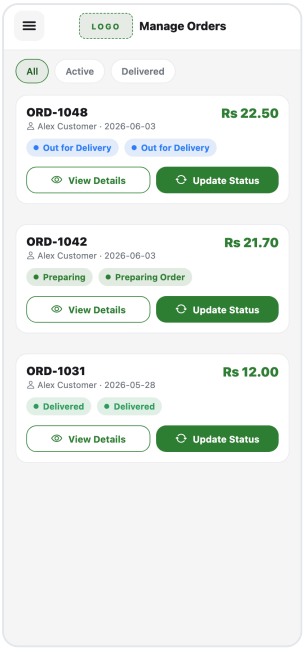
Screen	What it does
	**Checkout** — delivery address/phone, delivery date, time slot and payment method; validates required fields before placing the order.
	**My Orders** — Active vs Past tabs with order summary, **Reorder** and **Track Order**.

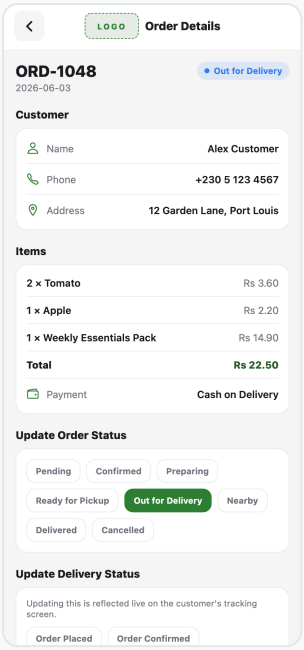
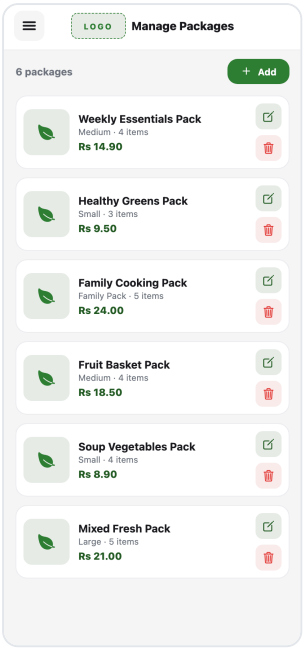
Screen	What it does
	**Delivery Tracking** — live status, map-style view with a moving driver marker, progress %, ETA, driver info and an auto-advancing timeline (pause/resume).
	**Profile** — editable customer details and saved addresses.

Screen	What it does
	**Settings** — pick any app colour (hue/shade/preset/hex) with live preview; persisted on device and applied app-wide.

Admin experience

Screen	What it does
	**Admin Dashboard** — business-intelligence panel: 12 KPI cards with trend + sparkline and a date-range filter driving the charts below.

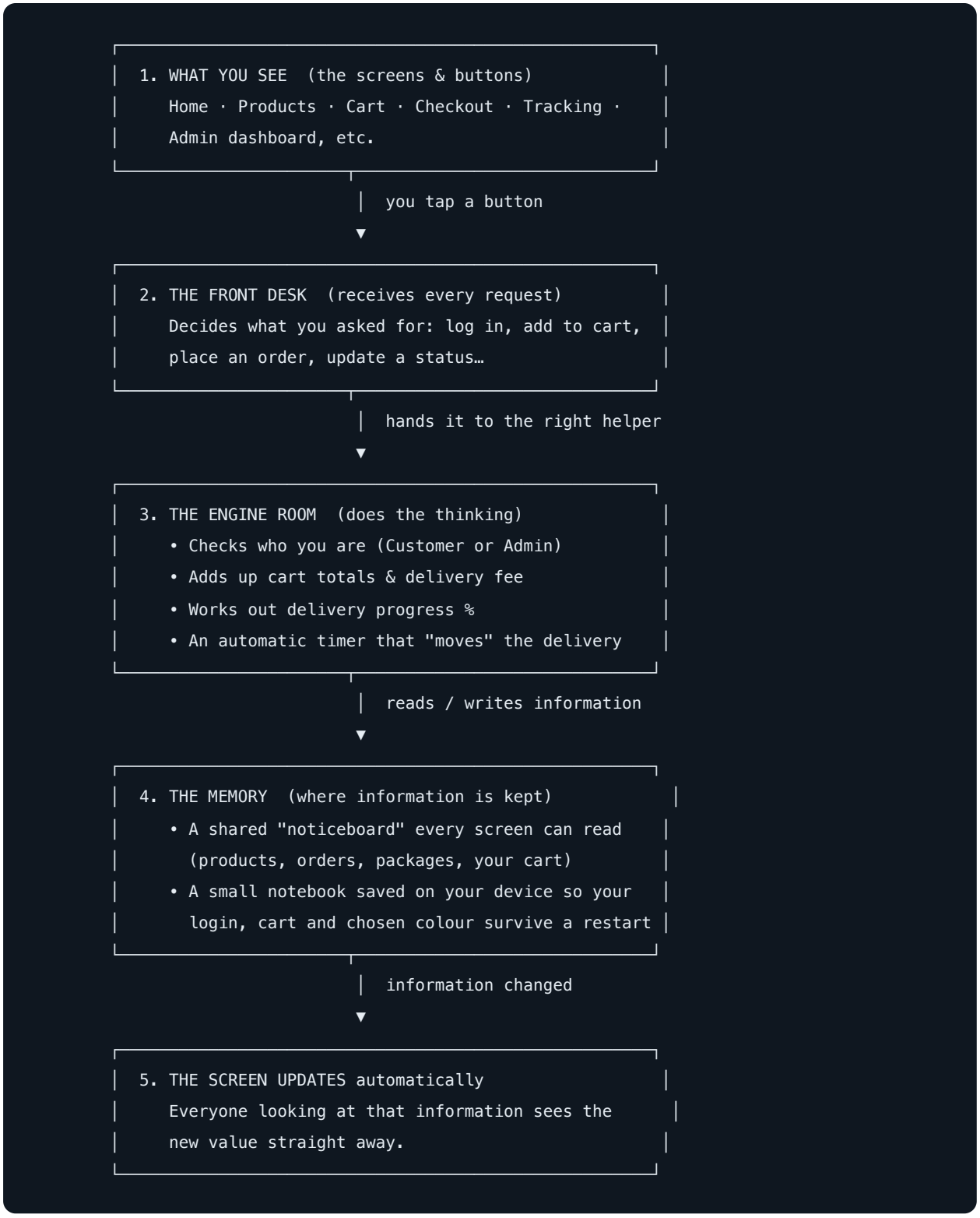
Screen	What it does
	**Manage Products** — list with category filter, stock badges, and add / edit / delete.
	**Manage Orders** — all orders with status chips, filter tabs, and quick status updates.

Screen	What it does
	**Order Details** — full order + item breakdown, and controls to **Update Order Status / Update Delivery Status** (reflected live on the customer's tracking screen).
	**Manage Packages** — ready-made packages with size/item counts and add / edit / delete.

3. How the App Works Behind the Scenes

Think of the app as a **small shop that keeps everything in its own back room** — there is no separate warehouse or head office on the internet. When you tap a button, the request never leaves your device; the app's own built-in "engine" handles it and updates the screen.

A simple picture



What each part means (in everyday words)

Part of the app	Plain-English meaning	Real-world comparison
The screens	The pages and buttons you interact with.	The shop floor.
The front desk	Catches your tap and figures out what you want.	A receptionist taking your request.

Part of the app	Plain-English meaning	Real-world comparison
The engine room	Does calculations and decisions (totals, who can see what, delivery progress).	The staff in the back room.
Login check	Confirms whether you're a Customer or an Admin and shows the right screens.	A name badge that unlocks certain doors.
The automatic timer	Every few seconds it nudges a delivery one step further along, all by itself.	A stopwatch that keeps the delivery "moving".
The shared noticeboard	One central place that holds products, orders and your cart, which every screen reads from.	A whiteboard everyone in the shop can see.
The device notebook	A tiny bit of storage on your phone so some things aren't forgotten when you close the app.	A sticky note left on the fridge.

Good to know

- **Nothing is sent over the internet.** No accounts, passwords, payments or personal data leave your device.
- **What is remembered vs. reset.** Your **login**, your **cart**, and your **chosen app colour** are saved on the device and survive a restart. The **product and order lists** start fresh from built-in sample data each time the app is fully reloaded.
- **Photos** you attach to a product (Admin) stay on the device only — they are not uploaded anywhere.
- **Not included** (because the app doesn't need them): an internet server, an online database, payment processing, or any AI service.

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4. The Most Important Journeys, Step by Step

These are the app's key features shown as simple, top-to-bottom journeys.

Journey A — Logging in

```
You tap "Login as Customer" (or Admin)
        ↓
The app checks which role you picked
        ↓
It remembers you on the device (so you stay logged in)
        ↓
It opens the right set of screens:
   Customer  →  shopping & tracking
   Admin     →  dashboard & management
```

Journey B — Placing an order

```
You fill in the Checkout form
(name, address, phone, delivery day, time slot, payment)
        ↓
The app checks the important fields are filled in
        │
        ├─ something missing → it shows a friendly error and stops
        │
        └─ all good ↓
The app creates your order (with an order number)
and empties your cart
        ↓
You see the "Order Confirmed" screen
        ↓
Your new order now appears in "My Orders" and can be tracked
```

Journey C — Live delivery tracking (runs by itself)

```
You open "Track Order"
        ↓
An automatic timer starts and, every few seconds…
        ↓
it moves the delivery to the next stage
(Order Placed → Confirmed → Preparing → Ready →
 Out for Delivery → Nearby → Delivered)
        ↓
On screen you see the driver icon move closer,
the progress bar fill up, and each step get ticked off
        ↓
It stops when the order is "Delivered"
(you can Pause or Resume at any time)
```

Journey D — Admin update shows up instantly for the customer

```
Admin opens an order and changes its delivery status
        ↓
The change is written to the shared "noticeboard"
        ↓
Because the customer's tracking screen reads from
the very same noticeboard…
        ↓
the customer sees the new status right away —
no refresh, no waiting
```

How to run it

```
npm install
npx expo start --web     # or press a / i for Android / iOS, or scan the QR in Expo Go
```

Log in as **Customer** or **Admin** (no password). Use **Logout** in the side menu to switch roles.